

# Introduction

❖ **Autacoids** from the Greek **autos** (“self”) and **akos** (“cure”) are endogenous organic molecules with potent pharmacologic effects

✓ not part of traditional immune or autonomic groups

## Classifications

✓ **Amine derived:** Histamine (amino acid: Histidine)  
Serotonin (Tryptophan)

✓ **Lipid Derivatives** (Platelet Activating Factor.  
Eicosanoids

✓ **Peptide derived:** Angiotensin, Bradykinin

# Lipid-Derived Autacoids: Eicosanoids and Platelet-Activating Factor

- ❖ Membrane lipids supply the substrate for the synthesis of
  - ✓ eicosanoids and
  - ✓ platelet-activating factor PAF
- ❖ Eicosanoids—arachidonate metabolites, prostaglandins (PGs), prostacyclin (PGI<sub>2</sub>), thromboxane A<sub>2</sub> (TxA<sub>2</sub>), leukotrienes (LTs), lipoxins, and hepoxilins—
  - ✓ not stored but are produced by most cells when a variety of physical, chemical, and hormonal stimuli activate **acyl hydrolases** that make arachidonate available
- ❖ **Eicosanoids and PAF** lipids contribute to inflammation, smooth muscle tone, hemostasis, thrombosis, parturition, and gastrointestinal secretion

# Eicosanoids

- ❖ Eicosanoids... ....eikosi ("twenty").
  - ✓ PGs, LTs, and related compounds
- ❖ Precursor essential fatty acids contain 20 carbons and 3, 4, or 5 double bonds.
  - ✓ **Arachidonic acid** (AA; 5,8,11,14-eicosatetraenoic acid) is the most abundant precursor, derived from dietary linoleic acid (9,12-octadecadienoic acid) or ingested directly as a dietary constituent

## BIOSYNTHESIS.

- **Biosynthesis of eicosanoids** is limited by the availability of AA and depends primarily on the removal of **esterified AA** from membrane **phospholipids or other** complex lipids by acyl hydrolases...phospholipase A<sub>2</sub> (PLA<sub>2</sub>).

# Eicosanoids

- Once liberated, AA is metabolized rapidly to oxygenated products..by cyclooxygenases (COXs), lipoxygenases (LOXs), and CYPs

# Lipoxygenase (lox) products.

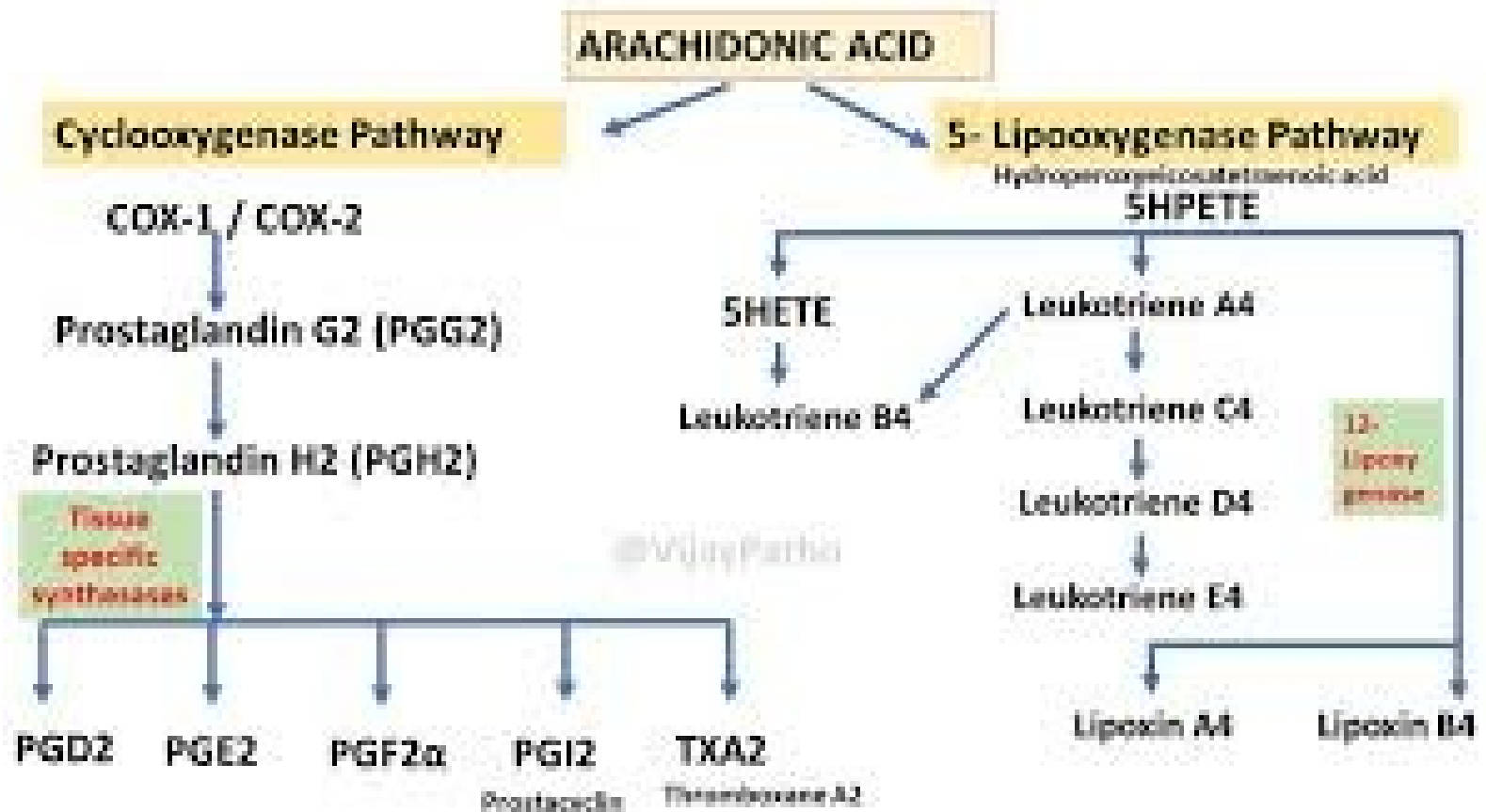


Figure: LOX and Cyclooxygenase pathways of arachidonic acid metabolism.

## Products of prostaglandin g/h synthases (cyclooxygenases)

❖ **PG endoperoxide G/H synthase** also is called **cyclooxygenase or COX**. Products of this pathway are

- PGs,
- prostacyclin (PGI<sub>2</sub>), and
- thromboxanes (TX<sub>2</sub>), collectively termed **prostanoids**.

**The pathway**

❖ Prostanoids are distinguished by substitutions on their **cyclopentane rings**, the number of **double bonds** in their side chains

- ❖ There are 2 distinct COX isoforms,
  - ✓ COX-1 and
  - ✓ COX-2.
- ❖ both enzymes may contribute to the generation of prostanoids of some physiologic and pathophysiologic processes.

### Expressions

**COX-1**, expressed constitutively in most cells, is the dominant source of prostanoids for **housekeeping functions**.

**COX-2**...upregulated by **cytokines**, shear stress, and growth factors and is the principal source of prostanoid formation in **inflammation and cancer**.

- ❖ These enzymes are expressed in a relatively cell-specific fashion. For example,
  - ✓ COX-1-derived TxA<sub>2</sub> is the dominant product in platelets, whereas
  - ✓ COX-2-derived PGE<sub>2</sub> and TxA<sub>2</sub> dominate in activated macrophages.

- ❖ Products of the LOX pathways are hydroxy fatty acid derivatives (HETEs),
  - LTs, and
  - lipoxins (LXs)
- ❖ LTs play a major role in the development and persistence of the inflammatory response.

# Inhibitors of eicosanoid biosynthesis

## ❖ Inhibition of PLA2

- decreases the release of the precursor fatty acid and the synthesis of all its metabolites.
- ❖ PLA2 may be inhibited by drugs that reduce the availability of  $\text{Ca}^{2+}$ .
- ❖ Glucocorticoids inhibit PLA2 indirectly by inducing **annexins** that modulate PLA2 activity.
- ❖ Glucocorticoids also downregulate induced expression of COX-2 but not of COX-1

# Inhibitors of eicosanoid biosynthesis

- ❖ Aspirin and other NSAIDs inhibit the COX
  - ✓ But do not inhibit **LOXs** and may cause increased formation of LTs by shunting of substrate to the **LOX pathway**.
- ❖ LTs may contribute to the GI side effects associated with NSAIDs.
- ❖ **COX-1** and **COX-2** differ in their sensitivity to inhibition by certain anti-inflammatory drugs.

## ❖ **selective inhibitors of the LOXs.**

- ✓ Zileuton, an inhibitor of 5-LOX, and
- ✓ selective LT-receptor antagonists (zafirlukast, pranlukast, and montelukast)
  - Used in the treatment of mild to moderate asthma

# Eicosanoid degradation

❖ Most eicosanoids are efficiently and rapidly inactivated.

**2 types of enzymatic catabolic reactions:**

**a rapid initial step**, catalyzed by widely distributed PG-specific enzymes,

**a second step** in which these metabolites are oxidized, probably by enzymes identical to those responsible for the  $\beta$  and  $\omega$  oxidation of fatty acids.

❖ Responsible organs for degradation (lung, kidney, and liver )

# Pharmacological effects

## Cardiovascular system

- In most vascular beds, PGE<sub>2</sub>, PGI<sub>2</sub>, and PGD<sub>2</sub> elicit vasodilation and a drop in blood pressure
- PGF<sub>2α</sub>.....constrictor of both pulmonary arteries and veins
  - ✓ does not alter blood pressure
- TxA<sub>2</sub> is a **potent vasoconstrictor** and a **mitogen** in smooth muscle cell

***Platelets.*** Mature platelets express only COX-1.

- TxA<sub>2</sub>, the major product of COX-1 in platelets,
  - ✓ induces platelet aggregation and amplifies the signal for other, more potent platelet agonists, such as [thrombin](#) and ADP.
- Both PGI<sub>2</sub> and PGD<sub>2</sub> inhibit the aggregation of platelets
- Low concentrations of PGE<sub>2</sub>.....enhance platelet aggregation.,
- higher concentrations of PGE<sub>2</sub>.....inhibit platelet aggregation.

# Inflammation and immunity

- ❖ LTs generally are pro-inflammatory and **lipoxins anti-inflammatory**.
- ❖ Prostanoids can exert both kinds of activity.
- ❖ COX-2 is the major source of prostanoids formed during and after an inflammatory response.
- ❖ **PGE<sub>2</sub>** and **PGI<sub>2</sub>** are the predominant pro-inflammatory. TxA<sub>2</sub> can increase platelet–leukocyte interaction.
- ❖ **PGD<sub>2</sub>**, also contribute to resolution of inflammation. PGs generally inhibit lymphocyte function and proliferation,
  - ✓ suppressing the immune response.

## **Bronchial and tracheal muscle**

- ❖ generally, **TxA<sub>2</sub>, PGF<sub>2α</sub>, and PGD<sub>2</sub> ....Contract**
  - ✓ PGE<sub>2</sub> and PGI<sub>2</sub>....Relax

**Uterus.....contracted by PGF<sub>2α</sub> and TxA<sub>2</sub> but are relaxed by PGI<sub>2</sub> and PGEs.**

## **Gastric and intestinal secretions.**

- ❖ **PGE<sub>2</sub> and PGI<sub>2</sub> contribute to increased mucus secretion (*cytoprotection*), reduced acid secretion, and reduced pepsin content.**
- ❖ **PGE<sub>2</sub> and its analogs also inhibit gastric damage caused by a variety of **ulcerogenic** agents and promote **healing of duodenal and gastric ulcers****

# GI smooth muscle

- ❖ **PGEs and PGFs** stimulate contraction of the main longitudinal muscle from **stomach to colon**  
✓ **Diarrhea, cramps.....in response to oral PGE. .**
- ❖ Circular muscle generally relaxes in response to **PGE2** and contracts in response to **PGF2 $\alpha$** .
- ❖ PGEs and PGFs stimulate the movement of water and electrolytes into the intestinal lumen.
- ❖ **PGE2** appears to contribute to the water and electrolyte loss in cholera

# Kidney

## ❖ Cortical COX-2-derived PGE<sub>2</sub> and PGI<sub>2</sub>

- increase renal blood flow and glomerular filtration
  - ✓ through their local vasodilating effects and inhibit tubular sodium reabsorption

## ❖ COX-1-derived products promote salt excretion in the collecting ducts.

*PAIN.* Inflammatory mediators, including LTs and PGs, increase the sensitivity of nociceptors and potentiate pain perception.

# Therapeutic uses

**Analogues and Inhibitors** ...nonselective NSAIDs, selective COX-2 inhibition, are used widely as anti-inflammatory drugs

- ✓ low-dose [aspirin](#) is employed frequently for cardio protective effect
- ✓ **LT antagonists** for treatment of asthma
- ✓ **PGF2 $\alpha$  agonists** are used in the treatment of open-angle glaucoma
- ✓ EP agonists to induce labor and to ameliorate gastric irritation owing to **NSAIDs**

# Therapeutic uses...

**Therapeutic Abortion.** PGEs, PGFs, and their analogs, are used to induce labor and terminate **pregnancy** by promoting **uterine contractions**

## **Drugs**

- ✓ Dinoprostone
- ✓ misoprostol with mifepristone/**Methotrexate**
- ✓ carboprost tromethamine

***Gastric Cytoprotection...*** PG analogs are used to suppress gastric ulceration.

- ✓ Misoprostol (Cytotec), a PGE1 analog, is approved for prevention of NSAID-induced gastric ulcers.

# Therapeutic uses...

**Impotence.....PGE1 (alprostadil)**

***Glaucoma.*** [Latanoprost](#), a  $\text{PGF}_{2\alpha}$  derivative, was the first prostanoid used for glaucoma.

- ✓ Similar prostanoids with this effect
  - **bimatoprost** and
  - **travoprost**

# Platelet-activating factor (PAF)

- ❖ PAF represents a family of phospholipids
- ❖ In human **neutrophils**, it consists predominantly of a mixture of the **16-** and **18-**carbon ethers, but its composition may change when cells are stimulated.
- ❖ PAF is not stored in cells but is synthesized from an **acyl precursor** in response to stimulation by a 2-step process

# PAF...

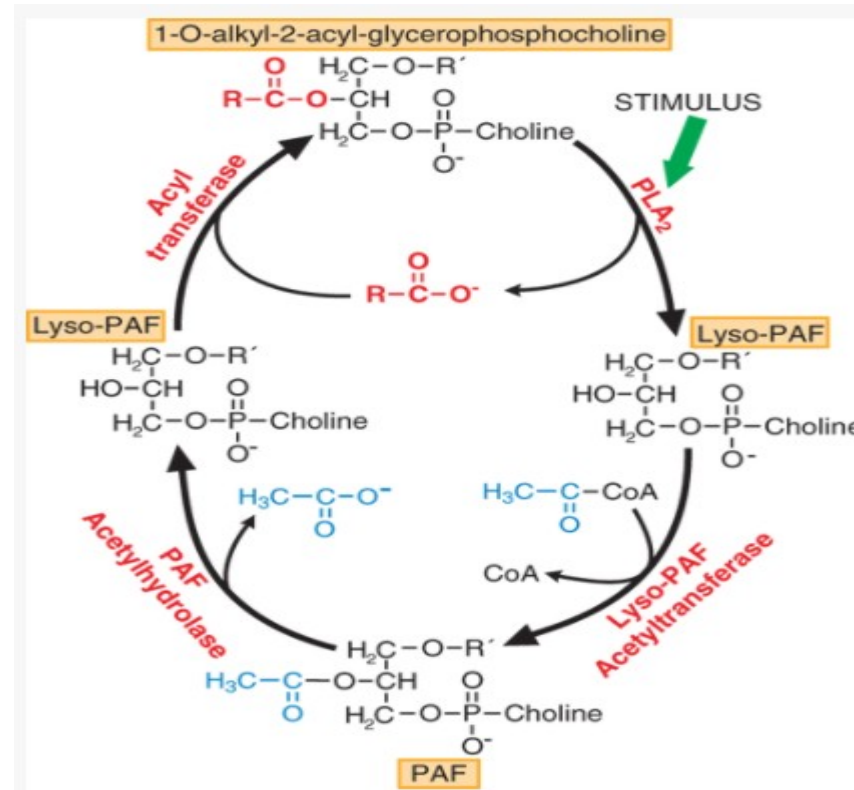


Figure: Synthesis and degradation of platelet-activating factor

# PAF...

**Synthesized** by platelets, neutrophils, monocytes, mast cells, eosinophils, renal mesangial cells, renal medullary cells, and vascular endothelial cells.

**Mechanism of action of PAF.** Extracellular PAF exerts its actions by stimulating a specific GPCR.

❖ The PAF receptor couples to Gq (to activate the PLC-IP<sub>3</sub>–Ca<sup>2+</sup> pathway) and to Gi (to inhibit adenylyl cyclase).

.

# Physiological and pathological functions of PAF

## **Inflammatory and Allergic Responses.**

- ✓ PAF administration reproduces many of the signs and symptoms in anaphylactic shock.
- ✓ In patients with asthma, **PAF antagonists** partially **inhibit the bronchoconstriction** induced by **antigen challenge** but not by challenges by **methacholine, exercise, or inhalation** of cold air.

# Cardiovascular System.

- ❖ IV PAF is a potent vasodilator in most vascular bed. This effects is independent of **effects on**
  - ✓ sympathetic innervation,
  - ✓ the renin–angiotensin system, or
  - ✓ AA metabolism and
- ❖ but likely results from a combination of **direct** and **indirect actions**. .
- ❖ **Intradermal injection** of PAF causes an **initial vasoconstriction** followed by a typical **wheal and flare**.
- ❖ PAF increases vascular permeability and edema

# Platelets.

- ❖ **IV injection of PAF causes formation of intravascular platelet aggregates and thrombocytopenia.**
- ❖ **Leukocytes**
  - ❖ **Activates polymorphonuclear leukocytes (eosinophils, neutrophils, and basophils).**
  - ❖ **Allows PMNs to aggregate, degranulate, and generate free radicals and LTs.**
  - ❖ **Enhance basophils to release histamine, activates mast cells, and induces cytokine release from monocytes. In addition, PAF promotes aggregation of monocytes and degranulation of eosinophils.**

# Smooth Muscle.

- ❖ It contracts GI, uterine, and pulmonary smooth muscle.
- ❖ It enhances the amplitude of spontaneous uterine contractions
- ❖ **PAF contracts airway smooth muscle.**
- ✓ aerosol, PAF increases airway resistance as well as the responsiveness to other bronchoconstrictors.
- ✓ **PAF↑mucus secretion**

# **Stomach.**

❖ **PAF is the most potent known ulcerogen.**

- ✓ IV PAF causes **hemorrhagic erosions** of the gastric mucosa that extend into **the submucosa**.

***Kidney.....decreases***

- ✓ renal blood flow,
- ✓ glomerular filtration rate,
- ✓ urine volume, and excretion of  $\text{Na}^+$

# Therapeutic Use

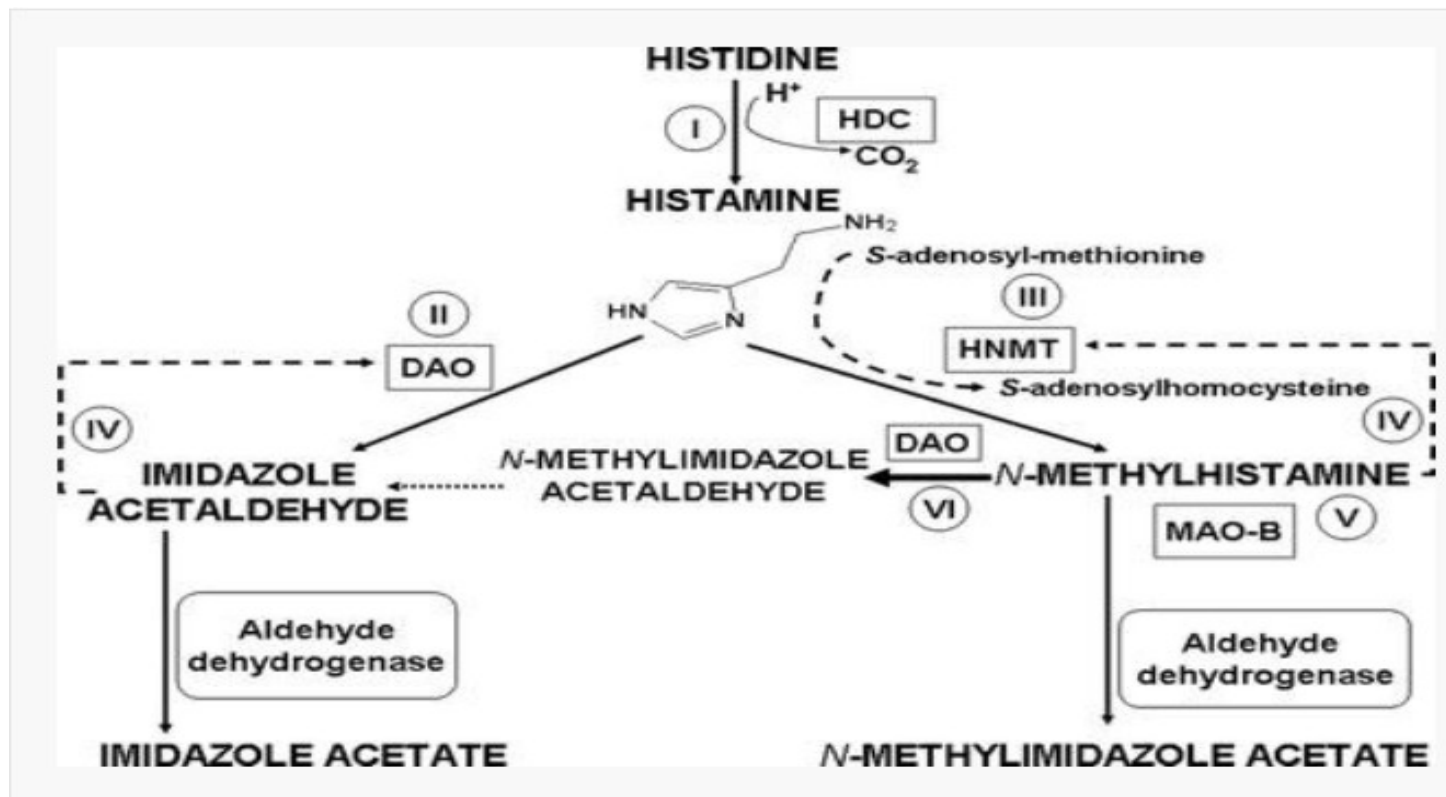
## ❖ **PAF Receptor Antagonists.**

- ✓ Several experimental PAF-receptor antagonists exist that selectively inhibit the actions of PAF in **vivo** and in **vitro**.
- ✓ **None has proven clinically useful**

# Histamine

**Histamine** is synthesized from the **amino acid histidine** and is

**Storage:** in vesicles in mast cells, enterochromaffin cells in the gut, some neurons, and a few other cell types.



# Receptors and Effects

- **H1 and H2, mediate most of the peripheral actions;**
- H3, H4 have also been identified
- ✓ The triple response, a classic demonstration of histamine effect on the **skin**, is mediated mainly by **H1 and H2 receptors**.

# Receptors and Effects

Some histamine and serotonin receptor subtypes.<sup>a</sup>

| Receptor Subtype      | Distribution                                              | Postreceptor Mechanisms                              | Prototypic Antagonists      |
|-----------------------|-----------------------------------------------------------|------------------------------------------------------|-----------------------------|
| H <sub>1</sub>        | Smooth muscle, endothelium, brain                         | G <sub>q</sub> : ↑ IP <sub>3</sub> , DAG             | Diphenhydramine, loratidine |
| H <sub>2</sub>        | Stomach, heart, mast cells, brain                         | G <sub>s</sub> : ↑ cAMP                              | Cimetidine                  |
| H <sub>3</sub>        | Nerve endings, CNS                                        | G <sub>i</sub> : ↓ cAMP                              | Clobenpropit <sup>b</sup>   |
| H <sub>4</sub>        | Leukocytes, CD4-T cells                                   | G <sub>i</sub> : ↓ cAMP                              | Thioperamide                |
| 5-HT <sub>1D/1B</sub> | Brain                                                     | G <sub>i</sub> : ↓ cAMP                              | —                           |
| 5-HT <sub>2</sub>     | Smooth muscle, platelets, brain                           | G <sub>q</sub> : ↑ IP <sub>3</sub> , DAG             | Ketanserin                  |
| 5-HT <sub>3</sub>     | Area postrema (CNS), sensory and enteric nerves           | Ligand-gated Na <sup>+</sup> /K <sup>+</sup> channel | Ondansetron                 |
| 5-HT <sub>4</sub>     | Presynaptic nerve terminals in the enteric nervous system | G <sub>s</sub> : ↑ cAMP                              | Tegaserod (partial agonist) |

## H1 receptor

❖ This is a Gq-coupled receptor diacylglycerol (DAG) are the second messengers.

❖ **responses includes**

✓ pain and itching in the skin,  
**bronchoconstriction, and vasodilation**

## H2 receptor

It is a Gs-coupled receptor mediates

- ✓ Allows gastric acid **secretion by parietal cells in the stomach.**
- ✓ has a **cardiac stimulant effect**
- ✓ reduce histamine **release from mast cells**—  
a negative feedback effect.

## H3 receptor

- ❖ It is a Gi-coupled receptor
- ❖ involved mainly in
  - ✓ presynaptic modulation of histaminergic neurotransmission **in the central nervous system (CNS).**
  - ✓ **Food intake and body weight increase in H3-receptor knockout animals**

## H4 receptor

- ❖ It is a Gi coupled receptor
- ❖ The H4 receptor is located on **leukocytes** (especially **eosinophils**) and mast cells and is involved in **chemotactic responses** by these cells..

### Clinical Use

- ❖ Histamine/analogues **has no therapeutic applications**, but drugs that block its effects at **H<sub>1</sub> and H<sub>2</sub> receptors** are very important in clinical medicine.
- ❖ No antagonists of H<sub>3</sub> or H<sub>4</sub> receptors are currently available for clinical use.

# H1 antagonists

- ❖ Two major subgroups or “generations” have been developed.
- ❖ Half-lives
  - ✓ older **H1 blockers** .....4 to 12 h.
  - ✓ **Second-generation agents** have half-lives of **12–24 h.**

## first-generation agents

- ❖ typified by **diphenhydramine**
- ❖ highly sedating agents with significant autonomic receptor-blocking effects.
- ❖ A newer subgroup of **first-generation** agents is less sedating and has much less autonomic effect.
  - ✓ Example **Chlorpheniramine** and **cyclizine** may be considered prototypes.

## **H1 antagonists**

- ✓ **The second-generation H1 blockers**
- ✓ , Example cetirizine, fexofenadine, and loratadine
  - less lipid soluble than the first-generation agents and have greatly reduced sedating and autonomic effects.

## Mechanism and Effects

**H1 blockers** are competitive pharmacologic antagonists or inverse agonists at the H1 receptor;

- ✓ Because their structure closely resembles that of muscarinic blockers and  $\alpha$ -adrenoceptor blockers,
  - ✓ many of the **first-generation** agents are potent **pharmacologic antagonists** at these autonomic receptors.
  - ✓ A few also block serotonin receptors.
- ✓ **H<sub>1</sub>-blocking drugs** have negligible effects at H<sub>2</sub> receptors.

# Therapeutic Uses

**H<sub>1</sub> blockers have major applications in allergies of the immediate type**

✓ hay fever and urticaria.

**anti-motion sickness**

✓ Diphenhydramine, dimenhydrinate, cyclizine, [meclizine](#), and promethazine

**chemotherapy-induced vomiting.**

✓ Diphenhydramine

**prevention of morning sickness in pregnancy**

**Doxylamine**, in combination with pyridoxine

# Therapeutic Uses

- ❖ Adverse effects of the first-generation H1 blockers.....exploited **therapeutically** (eg, in their use as hypnotics in over-the-counter sleep aids).

## Adverse Drug Reaction

**Sedation** is common, especially with

- ✓ diphenhydramine and
- ✓ promethazine
  - should not be consumed before operating machinery.
- ✓ **Sedation** is less common with second-generation agents, which do not enter the CNS readily.
- ✓ **Antimuscarinic effects** with some first-generation drugs in some patients.
- ✓ Alpha-adrenoceptor blockade,....significant with **phenothiazine** derivatives such as promethazine,
  - orthostatic hypotension.

# Drug Interaction

## Interactions occur between

- ❖ older antihistamines and other drugs with sedative effects (eg, benzodiazepines and alcohol).
- ❖ With drugs that inhibit hepatic metabolism....Increase effects
- For example, azole antifungal drugs and certain other CYP3A4 inhibitors
  - ✓ astemizole and terfenadine

# Histamine H2 antagonists

## Four H2 blockers are available;

- ✓ cimetidine is the prototype.
- ✓ Ranitidine,
- ✓ famotidine, and
- ✓ Nizatidine

- ❖ differ only in having fewer adverse effects than cimetidine.
- ❖ do not resemble H1 blockers structurally.
- ❖ **half-lives of 1–3 h.**
- ❖ **Duration.... 12–24 h**
- ❖ available in oral over-the-counter formulations.

# Mechanism and Effects

**MOA.....**They produce a surmountable pharmacologic blockade of histamine H<sub>2</sub> receptors.

- ✓ Relatively **selective** and (no significant effect at H<sub>1</sub> or autonomic receptors).
- ✓ The only **therapeutic effect** of clinical importance is the reduction of **gastric acid secretion**, but this is a very useful action. .

**Serotonin (5-hydroxytryptamine; 5-HT)**

**Serotonin** is produced from tryptophan and

- ✓ Serotonin plays a physiologic role as a neurotransmitter in both the **CNS** and the **enteric nervous**

**Storage....**stored in vesicles in **the enterochromaffin cells** of the **gut** and neurons of the **CNS** and **enteric nervous system**.

- ✓ Serotonin is also stored (but synthesized to only a minimal extent) **in platelets**
- ✓ After release back up into the **nerve ending** by a **serotonin reuptake** transporter (SERT).

# Metabolism

- ❖ After release, it is metabolized by monoamine oxidase.
- ❖ Excess production in the body can be detected by measuring its major metabolite
- ✓ **5-hydroxyindole acetic acid (5-HIAA)**, in the urine.

# Receptors and effects

## 5-HT1 receptors

- ✓ 5-HT1 receptors are Gi-protein-coupled.
- ✓ **5-HT1** receptors are most important in the brain and mediate synaptic inhibition via **increased potassium conductance**
- ✓ **Peripheral 5-HT1** receptors mediate both **excitatory** and **inhibitory effects** in various smooth muscle tissues.

## 5-HT<sub>2</sub> receptors

### It mediates

- ✓ **synaptic excitation** in the CNS and
- ✓ **smooth muscle contraction** (gut, bronchi, uterus, some vessels) or relaxation (other vessels).
- ✓ some of the **vasodilation, diarrhea, and bronchoconstriction** that occur as signs of carcinoid tumor
  - a neoplasm that releases serotonin and other substances.
- ✓ **5-HT<sub>2C</sub>** receptors mediate a **reduction** in appetite that has been used in the treatment of obesity.

## **5-HT<sub>3</sub> receptors**

- ✓ In the CNS, especially in the chemoreceptive area and vomiting center, and in peripheral sensory and enteric nerves.
  - ✓ Antagonists acting at this receptor are extremely useful antiemetic drugs.
- These receptors mediate excitation via a 5-HT-gated cation channel.

## **5-HT<sub>4</sub> receptors**

- ❖ 5-HT<sub>4</sub> receptors are found in the gastrointestinal tract and play an important role in intestinal motility.

## Clinical uses

**Serotonin** has no clinical applications, but more selective agonists are useful.

### **5-HT<sub>1D/1B</sub> agonists**

- ❖ Triptans..... Sumatriptan (prototype) and Naratriptan and
- ✓ first-line treatment for acute migraine and cluster headache attacks,.
  - These drugs are active orally and for nasal (sumatriptan)

### **5-HT<sub>2C</sub> agonists**

- **Lorcaserin** has been approved for the treatment of obesity.
- It activates receptors in the CNS and appears to moderately **reduce appetite**

## **5-HT4 Partial agonist**

**Tegaserod** is a newer drug that acts as an agonist in the colon.

- ✓ **approved** and briefly marketed for use in chronic **constipation**

- ✓ but because of cardiovascular toxicity, its use is now restricted.

## **Selective serotonin reuptake inhibitors (SSRI)**

- ✓ increase activity at central serotonergic synapses by inhibiting the serotonin reuptake transporter (SERT)

- ✓ **Adverse effects is associated with serotonin syndrome.**

## Serotonin antagonists

### Classification and Prototypes

#### 5-HT<sub>2</sub> blockers

- ✓ Ketanserin, **phenoxybenzamine (Irreversible blocker)**, and cyproheptadine (Compe) are effective.

These drugs are poorly selective for the receptor

**Other effect**----- $\alpha$ -blockade (ketanserin, phenoxybenzamine)

H<sub>1</sub>-blockade (cyproheptadine).

**5-HT<sub>3</sub> blockers**....Ondansetron, granisetron, dolasetron, and alosetron

**Toxicities**... diarrhea and headache

They have antiemetic actions in the area **postrema** of the medulla but

**alosetron** lacks this action

**Toxicities of alosetron**.....**Constipation**

# Clinical uses

- ❖ **Ketanserin** is used as an antihypertensive drug outside the **United States**.
- ❖ Ketanserin, cyproheptadine, and phenoxybenzamine may be of value (**separately or in combination**) for carcinoid tumor

## **Clinical uses...**

- ❖ **Ondansetron and its congeners** used in the control of vomiting associated with **cancer chemotherapy** and **postoperative vomiting**.
- ❖ **Alosetron** is used in the treatment of **irritable bowel syndrome associated with diarrhea**.